

AIRPORT PLANNING · GROUP PROJECT

Zaragoza Airport demand

Passenger forecast to 2050 and runway sizing based on Aena monthly data (2004–2026).

AUTHORS

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DATE

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Photo: Hugo González Duenas (JetPhotos)

DATA ANALYSED 268 months, Jan 2004 – Apr 2026	PASSENGERS IN 2025 707,493 base year	TRAFFIC–GDP ELASTICITY 1.07 78 quarters	PASSENGERS IN 2050 0.92–1.07_M three scenarios	DESIGN PEAK HOUR 444_{pax/h} August, base scenario	RUNWAYS REQUIRED 1 at 11–13 % of capacity
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01 Objective

Estimate how many passengers Zaragoza Airport will handle in 2050, how they will be spread over the year and what runway infrastructure the busiest hour will require.

The analysis starts from Aena's official monthly figures and links their evolution to the economy, measured by GDP. The result is not a single prediction but a plausible demand range within which the airport must operate.

AT A GLANCE

Series	268 months (244 without the pandemic)
Regression	78 quarters
Horizon	2050 (25 years from 2025)
Scenarios	GDP at 1.0 / 1.5 / 2.2 % a year

02 Data sources

All calculations are carried out in a working spreadsheet built from these sources. Each table and figure in this summary states its origin in a **SOURCE** line.

SOURCE	WHAT IT PROVIDES	PERIOD	WHERE IT IS USED
Aena · monthly passenger traffic	Departing, arriving and total passengers at the airport	Jan 2004 – Apr 2026	Whole series
Aena · monthly aircraft movements	Landings and take-offs	Jan 2023 – Apr 2026	Passengers per movement
Quarterly GDP index	Economic trend (included in the spreadsheet; no source or base year stated)	2004 Q3 – 2025 Q4	Elasticity and scenarios
Aena · 2025 annual report	Published annual total: 707,493 passengers, equal to the sum of the 12 months in the spreadsheet	2025	Cross-check
Press and Wikipedia	Totals for other years and later events (Ryanair, 2026)	2008 – 2026	Cross-check and context

SOURCE Working spreadsheet `evolucion_trafico_zaragoza.xlsx`; full references on page 6.

03 Spreadsheet structure

SHEET	CONTENT	ROLE
Comparativa mes año anterior	Full Aena series (268 months) and year-on-year change	Input data
Análisis tendencia	Series without 2020–2021 (244 months), moving average and seasonal index	Core calculation
Promedios · Media móvil total	Mean seasonal index per month; monthly and quarterly moving average	Seasonality and trend
PIB	Logarithms, elasticity, growth rate and 2050 scenarios	Forecast
HP · Operaciones	Passengers per month, day and peak hour; movements per hour and runways	Sizing

04 Method step by step

1 · Cleaning the series

The 24 months of 2020 and 2021 are removed: the pandemic produced anomalous values (4 passengers in April 2020) and disproportionate rebounds in 2021. They are not replaced by estimates; they are simply left out of all calculations, so the core analysis rests on 244 months.

2 · Trend: 12-month centred moving average

$$MA_t = (1/12) \cdot [0.5 \cdot X_{t-6} + X_{t-5} + \dots + X_{t+5} + 0.5 \cdot X_{t+6}]$$

X_t : passengers in month t . Every calendar month carries the same weight, which cancels out seasonality.

With an even window the centre falls between two months, so the end points carry half weight. Six months are lost at each end: the latest available trend is October 2025 (59,066 passengers a month, $\approx 708,800$ a year, consistent with the 707,493 of 2025).

3 · Seasonality: seasonal index

$$I_t = X_t / MA_t \cdot I_m = \text{mean of } I_t \text{ over all years for month } m \cdot \Sigma I_m \approx 12$$

Example: January 2025 had 44,040 passengers against a trend of 58,610.25, so $I = 44,040 / 58,610 = 0.75$: 25 % below an average month. Averaging every January gives the January index, 0.71.

MONTH	PASSENGERS	WEIGHT
Jul 2024	66,492	0.5
Aug 2024	71,136	1
Sep 2024	64,624	1
Oct 2024	61,961	1
Nov 2024	48,343	1
Dec 2024	54,624	1
Jan 2025	44,040	1
Feb 2025	43,859	1
Mar 2025	52,160	1
Apr 2025	67,643	1
May 2025	64,862	1
Jun 2025	61,293	1
Jul 2025	71,064	0.5
Moving average	58,610.25	

SOURCE Aena data, sheet “Análisis tendencia”.

4 · Elasticity of traffic with respect to GDP

Traffic T and GDP are assumed to follow a power law, $T = A \cdot GDP^\epsilon$. Taking logarithms gives a straight line whose slope is the elasticity:

$$\ln T = \ln A + \epsilon \cdot \ln GDP \rightarrow \epsilon = \text{SLOPE}(\ln T; \ln GDP)$$

Least squares over 78 quarters (quarterly moving average, 2020–2021 excluded).

Result: $\epsilon = 1.07$. For every 1 % of GDP growth, traffic grows by 1.07 %. The scatter is wide ($R^2 = 0.11$), so GDP explains only part of the variation in traffic.

The isolated point at the bottom right is 2025 Q4, which contains October only. Without it the slope would be 1.50 (note 1, page 6).

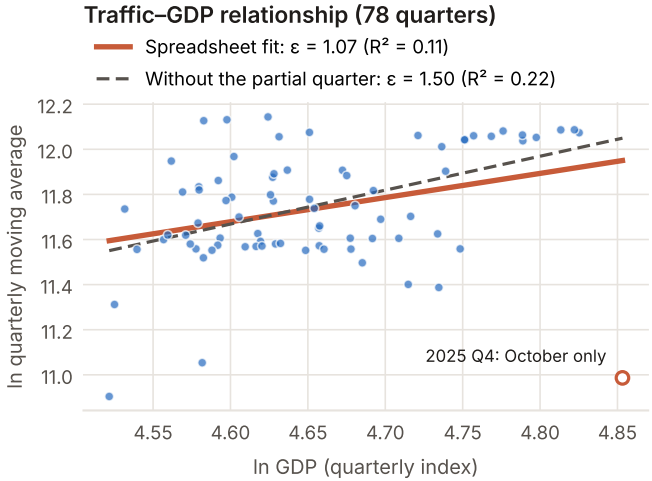


Figure 1 · Each point is a quarter. The orange line reproduces the spreadsheet elasticity.

SOURCE Sheet “PIB”; line recomputed from the same 78 points ($\epsilon = 1.0718$).

5 · Growth scenarios to 2050

The compound annual growth rate (CAGR) of the GDP index is used as a reference for the scenarios:

$$r = (V_{final} / V_{initial})^{1/t} - 1 = (128.15 / 91.99)^{1/21.5} - 1 = 1.55 \% \text{ a year}$$

86 quarters (21.5 years) between the first and last data points. It supports the 1.5 % base scenario.

$$i = \epsilon \cdot g \cdot PAX_{2050} = PAX_{2025} \cdot (1 + i)^{25}$$

g : scenario GDP growth; i : annual traffic growth; $PAX_{2025} = 707,493$ (sum of the 12 months of 2025).

SCENARIO	GDP, g	TRAFFIC, i	CALCULATION	PASSENGERS 2050
Pessimistic	1.0 %	1.07 %	$707,493 \cdot 1.0107^{25}$	923,583
Base	1.5 %	1.61 %	$707,493 \cdot 1.0161^{25}$	1,054,129
Optimistic	2.2 %	1.67 %*	$707,493 \cdot 1.0167^{25}$	1,069,258

SOURCE Sheet “PIB”. * The spreadsheet formula uses the historical 1.55 % instead of 2.2 % (note 2).

6 · From annual traffic to runways

$$MMT = PAX_{2050} \cdot I_m / 12 \rightarrow DMT = MMT / 31 \rightarrow HDP = DMT \cdot 0.11 \rightarrow mov/h = HDP / (100 \cdot 0.85) \rightarrow$$

$$runways = mov/h / 40$$

Peak month → average day → peak hour (11 % of the day) → movements (100-seat aircraft at 85 % load) → runways (40 movements per hour, the cautious end of the 40–45 range). Sheets “HP” and “Operaciones”.

05 Results: trend and seasonality

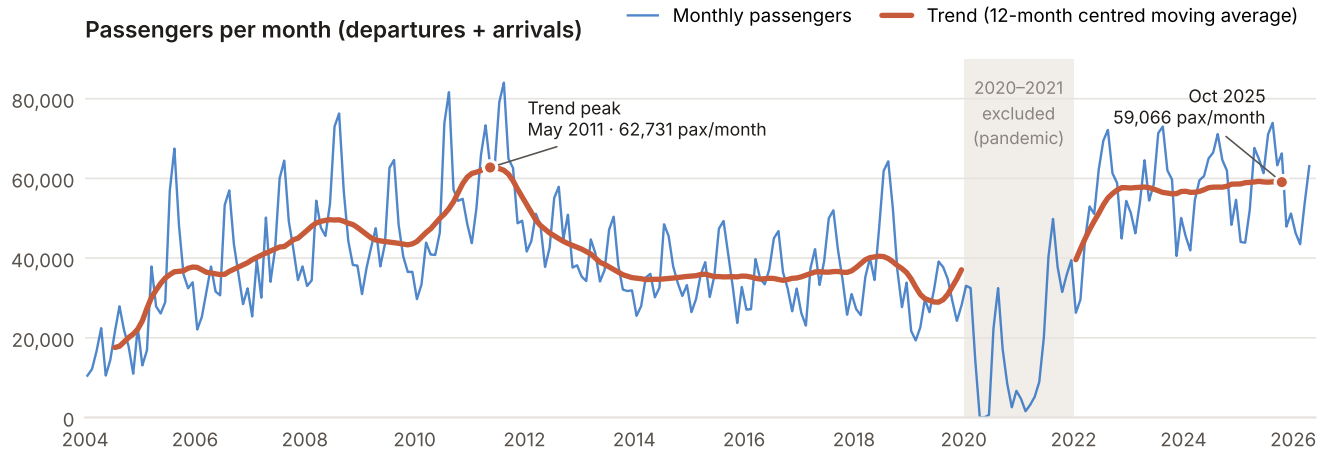


Figure 2 · Monthly passengers and trend. Peak in 2011, steady decline until 2019 and post-pandemic recovery to about 59,000 passengers a month.

SOURCE Aena (268 months, sheet “Comparativa mes año anterior”); trend from sheet “Análisis tendencia”.

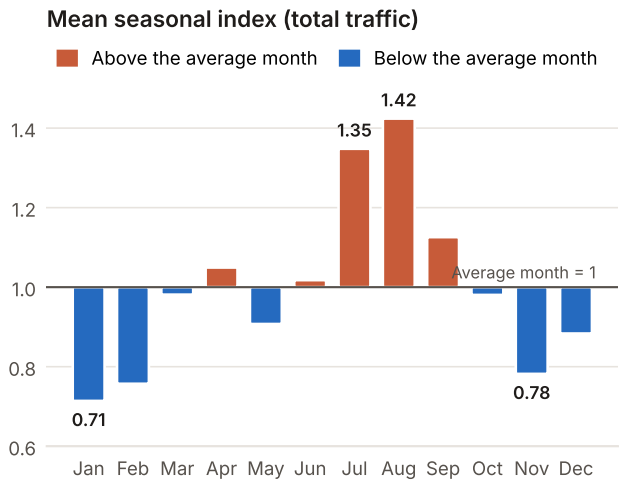


Figure 3 · August has 42 % more passengers than an average month; January, 29 % fewer.

MONTH	DEPARTURES	ARRIVALS	TOTAL
January	0.74	0.68	0.71
February	0.76	0.76	0.76
March	0.99	0.97	0.98
April	1.04	1.06	1.05
May	0.89	0.92	0.91
June	1.05	0.98	1.02
July	1.37	1.33	1.35
August	1.42	1.43	1.42
September	1.09	1.16	1.13
October	0.98	0.99	0.98
November	0.79	0.77	0.78
December	0.85	0.92	0.88
Sum	11.97	11.97	11.97

SOURCE Sheet “Promedios” (mean of each month's indices across all years).

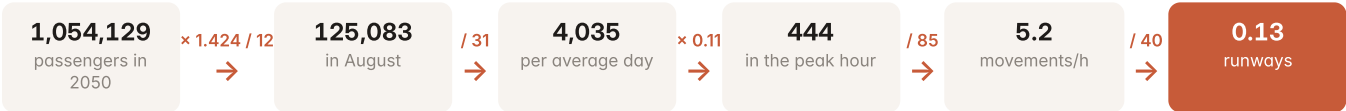
Departures and arrivals evolve almost identically (50.1 % and 49.9 % split), so the sizing uses the total index. August, with the highest index, is the **design month**.

06 Results: peak hour and runways in 2050

MONTH	INDEX	MMT BASE	DMT BASE	HDP PESSIM.	HDP BASE	HDP OPTIM.
January	0.71	62,715	2,023	195	223	226
February	0.76	66,502	2,145	207	236	239
March	0.98	86,192	2,780	268	306	310
April	1.05	92,181	2,974	287	327	332
May	0.91	79,706	2,571	248	283	287
June	1.02	89,395	2,884	278	317	322
July	1.35	118,416	3,820	368	420	426
August	1.42	125,083	4,035	389	444	450
September	1.13	98,897	3,190	307	351	356
October	0.98	86,129	2,778	268	306	310
November	0.78	68,686	2,216	214	244	247
December	0.88	77,603	2,503	241	275	279

SOURCE Sheet "HP". MMT: passengers in the month; DMT: passengers on an average day; HDP: passengers in the peak hour.

Design value calculation · August, base scenario



Movements and runways

QUANTITY	PESSIM.	BASE	OPTIM.
Peak - hour passengers	389	444	450
Movements per hour (/ 85)	4.6	5.2	5.3
Runways required (/ 40)	0.11	0.13	0.13

SOURCE Sheet "Operaciones".

In every scenario demand uses between 11 % and 13 % of the capacity of a single runway.

Cross-check: passengers per movement in 2025

MONTH	PASSENGERS	MOVEMENTS	PAX/MOV
January	44,040	674	65.3
April	67,643	930	72.7
July	71,064	932	76.2
August	73,934	849	87.1
October	66,268	1,081	61.3
Year 2025	707,493	10,508	67.3

SOURCE Aena, sheet "Operaciones". Average of the last 40 months: 67.4 pax/mov.

07 Notes on the data and calculations

#	NOTE	DETAIL	IMPACT
1	Partial quarter in the elasticity	The 2025 Q4 point contains October only and lowers the slope. Without it, $\epsilon = 1.50$ and the scenarios would give 1.03, 1.24 and 1.60 M passengers. <i>Own calculation.</i>	HIGH
2	Optimistic scenario	The formula uses the historical 1.55 % instead of the 2.2 % on the label. With 2.2 % the result would be $\approx 1,267,000$ passengers. <i>Own calculation.</i>	HIGH
3	Peak-hour parameters	The 11 % factor has no source; 85 pax/mov is used against 67.4 observed (with it, 5.8–6.7 mov/h). The conclusion does not change. <i>Own calculation.</i>	MEDIUM
4	2019 data	The spreadsheet total (344,978) is close to the published international passengers (345,301), not to the total (467,774).	MEDIUM
5	GDP index	No source, scope or base year; it does not appear to be seasonally adjusted.	MEDIUM
6	Joining 2019 and 2022	Ten moving averages mix months from both years; effect on the indices < 0.01 . <i>Own calculation.</i>	LOW
7	Year-on-year change	The column on the first sheet shows some decreases as increases.	LOW

08 Conclusions

ONE RUNWAY IS ENOUGH IN EVERY SCENARIO

The 2050 peak hour requires between 4.6 and 5.3 movements per hour. Even correcting notes 1 to 3 at once (≈ 1.6 M passengers and 67.4 pax/mov) would give about 10 movements per hour, a quarter of the runway's capacity. *Own calculation.*

AFTER THE SPREADSHEET

In September 2025 Ryanair announced a 45 % capacity cut at Zaragoza, and August 2026 recorded 63,717 passengers (-13.8 % year on year). Since the scenarios start from 2025, they should be reviewed with the 2026 data.

09 References

- Aena. (2026). *El Aeropuerto de Zaragoza cierra 2025 con más de 707.000 pasajeros*. aena.es
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- Aragón Digital. (2020, 13 January). *El Aeropuerto de Zaragoza bate el récord de tráfico de mercancías con más de 180.000 toneladas*.
- Aragón Digital. (2025, 3 September). *Ryanair recorta un 45 % su capacidad en Zaragoza tras la subida de tasas de Aena*.
- El Español. (2026, 23 September). *El Aeropuerto de Zaragoza pierde 10.000 pasajeros en agosto respecto a 2025*.
- Wikipedia. (n.d.). *Aeropuerto de Zaragoza · Zaragoza Airport*.

FILES data/evolucion_trafico_zaragoza.xlsx (working spreadsheet) and Informe_demanda_aeropuerto_Zaragoza.pdf (full report, in Spanish), downloadable from the portfolio.